Hal Hoffmeyer

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EDUCATION

University of Michigan | B.S.E. Computer Science | GPA: 3.97 May 2027

Coursework: Distributed Systems, Machine Learning, Web Systems, Computer Organization, Data Structures and Algorithms, Discrete Math, Introduction to Probability, Applied Linear Algebra, Statistics and Data Analysis

Awards: James B. Angell Scholar, William J. Branstrom Freshman Prize, University Honors

EXPERIENCE

Cleveland-Cliffs | Software Engineer Intern May 2026 – August 2026

- Built 5+ ASP.NET web applications to manage lab equipment organization and logistics for smaller steel mills.
- Integrated the Web Bluetooth API and ZPL to print QR codes directly from web forms to local printers.
- Programmed a C background service to detect temperature inversions from sensor data, inserting compressed datapoints into tables to trigger alarms that report possible steel rusting.

Zeta Pi Tech Fraternity | Full-Stack Web Developer June 2025 — Present

- Created ZProfile, a full-stack Next.js web app for 100+ users designed to streamline management and logistics.
- Implemented an attendance tracker and event creator integrated with Supabase using a custom dual list box component and React Hook Form.
- Built profile page allowing students to log course history by pulling course information from U-M API and writing custom multiselect component to store student course data into Supabase.
- Developed an interactive course directory by querying previously stored Supabase data and built logic for search, debounce, and filtering features using React hooks and PostgreSQL.
- Implemented a bug reporting system leveraging the GitHub REST API, automatically creating GitHub issues from user submissions to streamline bug tracking and improve response time.

Detroit Techno Film Web Platform | Web Developer September 2024 — October 2025

- Developed a companion web app for a documentary, using Next.js to showcase dynamic music and video dashboards, NextAuth.js for secure user authentication, AWS S3 for video storage, and Vercel for database.
- Led developer meetings to establish a unified Git workflow amongst team members and merge back-end with front-end components.

PROJECTS

MNIST Classifier | *C++* March 2026

- Developed a neural network from scratch for MNIST digit classification, achieving 98% accuracy using ReLU and Softmax activation functions, optimized forward and back propagation, and hyperparameter sweeping.

ICU Patient Mortality Prediction | *Python, NumPy, Scikit-learn, Pandas* February 2026

- Developed a Logistic Regression classifier to predict ICU patient mortality, implementing L1/L2 regularization and extensive hyperparameter sweeps to optimize model selection.
- Engineered a high-dimensional feature pipeline from irregular ICU time-series data, utilizing 5-fold cross-validation and bootstrap sampling to achieve an AUROC of 0.85 on unseen clinical test data.

Sharded Key/Value Store with Paxos Groups | *Go* April 2026

- Built a fault-tolerant, sharded key/value store in Go with shard migration across Paxos replicated groups.
- Implemented a shard master for dynamic shard rebalancing, atomic state handoffs, and duplicate request detection. Ensured linearizability by serializing all operations in the Paxos consensus log.

Minesweeper | *React, TypeScript, Tailwind* December 2024

- Developed a Minesweeper game, integrating game logic while handling state management and dynamic grid rendering with React components and event listeners in TypeScript for a seamless user experience.

Languages: C/C++, JavaScript/TypeScript, HTML/CSS, SQL, C#, Go, Python, R, Java

Tools: PyTorch, NumPy, Flask, UNIX, Makefile, React, Next.js, jQuery, .NET, PostgreSQL, Supabase, Git, SVN, Jira, Vim

Certifications: N3 Japanese Language Proficiency Test